

Shankar Karuppayah

Curriculum vitae

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Employment History

- Oct'24–present **Associate Professor**,
Cybersecurity Research Centre (CYRES),
Universiti Sains Malaysia, Malaysia
www.cyres.usm.my
- 2023–present **Deputy Director**,
Cybersecurity Research Centre (CYRES),
Universiti Sains Malaysia, Malaysia
www.cyres.usm.my
- Oct'23–present **CEO & Founder**,
BitRanger Sdn Bhd,
USM Cybersecurity Spinoff, Malaysia
www.bitranger.io
- 2016–Sept'24 **Senior Lecturer**,
National Advanced IPv6 Centre (NAv6),
Universiti Sains Malaysia, Malaysia
www.nav6.usm.my
- 2020–2021 **Area Head**,
Telecooperation Lab (TK),
Technische Universität Darmstadt, Germany
www.tk.tu-darmstadt.de
- 2019–2021 **Postdoctoral Researcher**,
Telecooperation Lab (TK),
Technische Universität Darmstadt, Germany
www.tk.tu-darmstadt.de

Education

- 2012–2016 **PhD, Cyber Security**, *Technische Universität Darmstadt*, Darmstadt, Germany
Thesis title: Advanced Monitoring of P2P Botnets
Supervisor: Prof. Dr. Max Mühlhäuser
- 2009–2011 **MSc, Software Systems Engineering**, *King Mongkut's Univ. of Tech. North Bangkok*, Bangkok, Thailand
Curriculum based on the syllabus of RWTH Aachen, Germany
- 2006–2009 **BSc (Hons), Computer Science**, *Universiti Sains Malaysia*, Penang, Malaysia

Membership and Professional Activities

Membership

- Asia Pacific Advanced Network (APAN) - Security Working Group
Chair
- Institute of Electrical and Electronics Engineers (IEEE)
Member
- Malaysian Research and Education Network (MyREN)
Internet Security Working Group Deputy Head
- Malaysian Board of Technologists (MBOT)
Professional Technologist

Journal Reviewer

- ACM Computing Surveys (CSUR)
- IEEE Transactions on Information Forensics & Security (T-IFS)
- IEEE Journal on Selected Areas in Communications(JSAC)
- PeerJ Computer Science
- Journal of Information Security and Applications (JISA)
- Computers and Electrical Engineering

Committees

- MRANTI Cybersecurity Technical Advisory Panel, AV Living Lab
- USM Mobile Access Coordinator (COVID-19)
- USM Industry Liaison Fellow
- USM Cyber Security Awareness Program - Subject Matter Expert

(Selected) Consultancy Projects

2025 **OpenSOC SEA-NREN Initiative: Empowering Research Networks Against Cyber Threats**

This APNIC ISIF project aims to support emerging National Research and Education Networks (NRENs) in establishing Security Operations Centre (SOC) capabilities using open-source technologies. The project focuses on building a collaborative framework that enables participating networks to monitor threats, share threat intelligence, and strengthen cybersecurity resilience across regional research and education infrastructures.

Consultancy Project (Principal Consultant)

2024 **Cybersecurity Awareness Training**

A full-day awareness training program crafted to upskill members of an academic institution's in Kuala Lumpur, Malaysia.

Consultancy Project (Trainer)

2023 **Embedded Systems Upskilling Program For Computer Science/Engineering Graduates**

A three-months program crafted to upskill existing industry engineers with knowledge on embedded systems that is highly in demand within the manufacturing industries in Penang, Malaysia.

Industry Grant (Trainer)

- 2021 **Cyber-Physical Security Design for Energy Aggregation: Applicability In The Malaysian Landscape**
A study and design on cyber-physical security design for energy aggregation domains and its application on the Malaysian market
International Grant (Co-Researcher)
- 2021 **Joint Solution Design To The Technical Problems Faced By TF-AMD**
A project to study and propose designs as solution to the technical problems faced by TF-AMD
Industry Grant (Co-Researcher)
- 2018 **Development of Laboratory Information Management System**
Design and development of a customized web-based LIMS system to facilitate the day-to-day operations of a service-based research centre within USM
Internal Project (Project Leader)
- 2017 **IoT-Based Recipe Management System**
Design and development of a in-house system to interconnect various equipment in a shop floor with recipe management and data analytics capabilities
Industry Grant (Co-Researcher)

(Selected) Research Projects

- 2025 **Research Into Security Layer and Communication for i-Connect Industry 4.0 in Manufacturing Modular Model Industry 4.0 Digital Twin Factory With Intelligence System on Systems and IOT Connectivity for Industry 4.0**
An industrial research project to design and propose a lightweight AI-powered IDS mechanism for Manufacturing Industry 4.0 environment
Industry Grant (Co-Researcher)
- 2023 **Implementation and Experiment of the Research Platform in Malaysia For Real-Time Malicious TLS Traffic Detection Using Machine Learning Classifier**
An international collaborative research project to design a research platform to detect TLS-based malware communication using machine learning classifiers
International Grant (Project Leader)
- 2023 **Designing and Developing A Security Operations Center As A Service Sensor Prototype**
An industry project to design and develop a sensor to be leveraged for offering a Security Operation Center as a service (SOCaaS)
Industry (Project Leader)
- 2022 **User-mobility Optimized Routing Protocol For Enabling Autonomous And Self-managing Disaster Communication Network**
An TRGS research grant to design and implement a user-mobility optimized routing protocol specific for disaster management scenarios.
National Grant (Project Leader)
- 2021 **A periodicity-agnostic method for detecting parasitic botnets in Distributed Hash Table-based peer-to-peer networks**
An FRGS research grant to study and propose methods to detect parasitic botnets in DHT-based P2P network
National Grant (Project Leader)

2019 Novel P2P Botnet Detection and Endpoint Mobilization

A Royal Bank of Canada (RBC) funded project aimed at carrying out research on advanced P2P botnet detection and countermeasures

International Industry Grant (Associate Researcher - TU Darmstadt)

(Selected) Publications

- Federated Learning/IDS Tasneem Al-Ghadi, Selvakumar Minickam, I Dewa Made Widia, Eka Ratri Noor Wulandari, **Shankar Karuppayah**, Leveraging Federated Learning for DoS Attack Detection in IoT Networks Based on Ensemble Feature Selection and Deep Learning Models, *Cyber Security and Applications*, 2025,  doi:10.1016/j.csa.2025.100098
- Privacy August See, Mathias Fischer, Maximilian Gehring, **Shankar Karuppayah**, Binary Sight-Seeing: Accelerating Reverse Engineering via Point-of-Interest-Beacons, *ACSAC* 2023,  doi:10.1016/j.clsr.2021.105652
- Malware Leon Böck, Martin Fejrskov, Katerina Demetzou, **Shankar Karuppayah**, Max Mühlhäuser, Emmanouil Vasilomanolakis, Processing of botnet tracking data under the GDPR, *Computer Law & Security Review*, 2022,  doi:10.1145/3627106.3627139
- SDN Abdijalil Abdullahi, Selvakumar Manickam, **Shankar Karuppayah**, A Review of Scalability Issues in SoftwareDefined Exchange Point (SDX) Approaches: State-of-the-Art, *IEEE Access*, **9**, 74499–74509 (2021).  doi:10.1109/ACCESS.2021.3069808
- SECS/GEM Ashish Jaisan, Selvakumar Manickam, Shams A. Laghari, Shafiq Ul Rehman, **Shankar Karuppayah**, Secured SECS/GEM: A Security Mechanism for M2M Communication in Industry 4.0 Ecosystem, *International Journal of Advanced Computer Science and Applications*, **12/8**, 241–250 (2021).  doi:10.14569/IJACSA.2021.0120828
- IOT Ahmed J. Hintaw and Selvakumar Manickam and **Shankar Karuppayah**, Mohammed Faiz Aboalmaaly, A Brief Review on MQTT's Security Issues within the Internet of Things (IoT), *Journal of Communications* **6**, 463–469 (2019).  doi:10.12720/jcm.14.6.463469
- IOT/IDS Kandhro I.A., Alanazi S.M., Ali F., Kehar A., Fatima K., Uddin M., **Shankar Karuppayah**, Detection of Real-Time Malicious Intrusions and Attacks in IoT Empowered Cybersecurity Infrastructures, *IEEE Access*, (2023)  doi:10.1109/ACCESS.2023.3238664
- IDS Emmanouil Vasilomanolakis, **Shankar Karuppayah**, Max Mühlhäuser, Mathias Fischer, Taxonomy and Survey of Collaborative Intrusion Detection, *ACM Computing Surveys*, (2015)  doi:10.1145/2716260
- P2P Botnets Kabla A.H.H., Thamrin A.H., Mohammed F.R. Anbar, Selvakumar A/L Manickam, **Shankar Karuppayah**, PeerAmbush: Multi-Layer Perceptron to Detect Peer-to-Peer Botnet, *SYMMETRY*, (2022)  doi:10.3390/sym14122483
- P2P Botnets **Shankar Karuppayah**, Advanced Monitoring in P2P Botnets: A Dual Perspective, *Springer*, (2018)  doi:10.1007/9789811090509
- P2P Botnets Leon Böck, Emmanouil Vasilomanolakis, Max Mühlhäuser, **Shankar Karuppayah**, Next Generation P2P Botnets: Monitoring Under Adverse Conditions, *Research in Attacks, Intrusions, and Defenses - 21st International Symposium (RAID)*, (2018)  doi:10.1007/9783030004705_24

h-index  (Google Scholar): 24  (Scopus): 19  (Web of Science): 12

Language Skills

English Proficient
Malay Proficient
Deutsch Beginner
Tamil Conversational

IELTS Band 8.0
National Language
EU-CEFR Level A2
Mother Tongue